Station Record **LT 4.4 · Four tools, two stories**

Code name _____ Per ___

CATCH SOMEONE

1 PCR · pipe cleaners

Cycles	0	1	2	3
DNA molecules	1			

The PCR machine heats and cools over and over because _____ _____, and each cycle _____ the DNA.

2 Crime Lab gel · iPad Case number _____



Copy the gel band for band. Label the suspects' roles under 1, 2, 3.

1 _____

2 _____

3 _____

Claim: the DNA at the scene came from _____

Evidence:

_____

_____

Reasoning:

_____

_____

_____

Does a match prove this person did it? yes / no

TREAT SOMEONE

3 Plasmid · insulin · POP beads

Number the steps 1–4 in order.

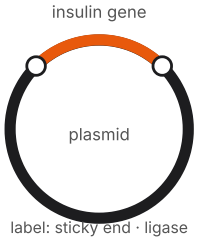
___ Ligase seals the gene into the plasmid.

___ The bacterium reads the gene and makes insulin.

___ A restriction enzyme cuts the insulin gene out and opens the plasmid.

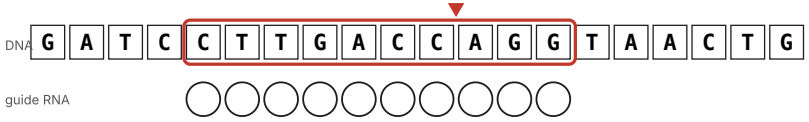
___ A bacterium takes in the recombinant plasmid.

Beads in the new ring: _____



4 CRISPR · sickle cell · iPad

Write the guide RNA under the boxed DNA, one base per circle (U for T). Draw a line through the strip where Cas9 cuts, at the ▼.



After the cut, the patient's cells turn _____ back on, and the red blood cells stay _____.

Which Tool? · after the stations

Sort the 8 cards at your table onto PCR, gel, plasmid, CRISPR (or "no tool here"). Write one card that needed two tools, and the two tools in order:

_____

_____

On the iPad, today's Start Here: Station Check (2 questions per station) · then the Trojan Gene reading and CER.

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